Business Requirements Document

AI Front-Desk Agent for Solo Clinics — MVP (Phase 1, Single Test Client)

Product	Clinic Appointment Intake & Logistics Follow-Up Agent
Document Version	1.0
Status	Draft — MVP scope, single test client, build/test environment only
Prepared For	Clinic Owner / Practice Manager
Delivery Vehicle	Google Apps Script + Google Forms/Sheets + Twilio + Claude API

IMPORTANT — READ BEFORE USE WITH REAL PATIENTS: this MVP is a build/test environment. Patient name + phone + "has an appointment" is Protected Health Information (PHI) under HIPAA regardless of how non-clinical the stored fields are. Connecting this system to real patients requires signed HIPAA Business Associate Agreements (BAAs) with Google Workspace, Twilio, and Anthropic, plus the practice's own compliance/legal review. This document is not legal or compliance advice.

1. Executive Summary

Solo clinics lose patient goodwill and staff time to slow, manual appointment-logistics handling: confirming new-patient intake, sending reminders, and answering routine scheduling questions all compete with in-room care for staff attention. This project delivers an AI-assisted front-desk system that captures appointment requests from a web form, confirms them by SMS within 60 seconds, sends reminders automatically, and drafts logistics-only replies for staff to review and approve — with a hard, structural boundary that keeps clinical/diagnostic content out of the AI path entirely.

2. Business Objectives

- Reduce appointment-logistics response time from hours to under 60 seconds.
- Cut staff time spent on routine scheduling confirmations and reminders.
- Guarantee that clinical/diagnostic content never reaches the AI system, by design, not by policy alone.
- Give staff a single dashboard view of intake status, last contact, and any items needing a human.
- Prove the model end-to-end for one test (non-real) patient before considering real-patient use, which requires BAAs first.

3. Scope

3.1 In Scope (Phase 1 MVP)

- Web-embeddable Google Form capturing name, phone, reason for visit (closed non-clinical categories only), and preferred appointment window.

- Real-time write of form submissions to a dedicated Google Sheet (no shared spreadsheet across clients).
- Automated first-contact SMS within 60 seconds of form submission.
- A two-step reminder sequence (24h, 72h) for appointments that have not been confirmed.
- Inbound SMS reply handling: a local keyword pre-filter runs before any text reaches the Claude API; only messages that pass the filter are sent to the model, scoped strictly to appointment logistics.
- Hard, no-AI-involvement escalation to clinic staff for anything that reads as clinical: symptoms, diagnosis, medication, test results, or a direct request for a human/nurse/doctor.
- A staff approval queue (web dashboard) where a logistics-only AI draft is reviewed and explicitly sent.
- A dashboard showing intake status, last contact time, next scheduled action, and average first-response time — with no clinical data displayed, by design.

3.2 Out of Scope (Phase 1)

- Any AI-generated clinical, diagnostic, or treatment content, under any circumstance.
- Real-patient PHI processing until Google Workspace, Twilio, and Anthropic BAAs are executed.
- Auto-sending AI-drafted replies without staff approval.
- Multi-tenant infrastructure (Phase 1 targets one test client).
- EHR/PMS integration, insurance verification, or billing workflows.

4. Stakeholders

Role	Interest
Clinic Owner / Practice Manager	Primary user; approves AI drafts; owns Twilio/Claude accounts and BAA decisions.
Front-desk / Clinical Staff	Receives escalation alerts; handles anything flagged as clinical directly.
Patient	Receives appointment confirmation, reminders, and (once approved) logistics-only AI-assisted replies.
Solution Implementer	Deploys and configures the Apps Script project, Form, Sheet, and Twilio number per client.

5. Functional Requirements

5.1 Intake Layer

ID	Requirement
FR-1	The system shall present a Google Form with fields: name, phone, reason for visit, preferred appointment window.
FR-2	"Reason for visit" shall be a closed dropdown (New Patient / Follow-up / General Question / Prescription Refill Request / Other) and shall never be a free-text symptom field.
FR-3	The system shall write each submission to a dedicated Google Sheet in real time via the native Forms→Sheets integration.

5.2 Trigger & Orchestration Layer

ID	Requirement
FR-4	On new form submission, the system shall format and send a first-contact confirmation SMS within 60 seconds.
FR-5	The system shall log contact status (contacted / pending / booked / escalated / closed) back to the Sheet.
FR-6	The system shall run a recurring reminder sweep (default: every 15 minutes) for appointments not yet confirmed.
FR-7	The system shall stop the reminder sequence once the patient replies, the appointment is confirmed, or the row is escalated.

5.3 Phone/Messaging Layer — Clinical-Data Boundary

ID	Requirement
FR-8	The system shall send outbound SMS via Twilio using per-client credentials.
FR-9	Every inbound patient message shall be checked against a local keyword/pattern filter (symptoms, diagnosis, medication, test results, mental health terms, emergency terms, etc.) before any call is made to the Claude API.
FR-10	If the local filter flags a message, the system shall skip the LLM call entirely and escalate directly to clinic staff by SMS — the model shall never see flagged text.
FR-11	If the message passes the local filter, the system shall route it to a Claude API call scoped by a system prompt to appointment logistics only, which itself refuses and escalates anything clinical.
FR-12	The system shall never send an AI-drafted reply to a patient automatically; every draft requires explicit staff approval.
FR-13	The raw conversation log (needed operationally for staff to call the patient back) may contain patient-submitted text verbatim, but the AI system itself shall never generate, interpret, or act on clinical content.

5.4 Dashboard Layer

ID	Requirement
FR-14	The system shall provide a web dashboard showing, per patient: status, last contact time, next scheduled action, and response-time metric — logistics fields only.
FR-15	The dashboard shall clearly flag escalated rows as "call patient, no AI draft" rather than showing any AI-generated content for them.
FR-16	The dashboard shall be restricted to authorized staff Google accounts, separate from the public Twilio webhook endpoint.

6. Non-Functional Requirements

Category	Requirement
Performance	First-contact SMS shall be sent within 60 seconds of form submission under normal Apps Script quota conditions.

Category	Requirement
Compliance	Real-patient use requires executed BAAs with Google Workspace, Twilio, and Anthropic; until then this is a build/test environment only, per Section 0.
Security — Isolation	Each client shall have a fully separate Spreadsheet and Apps Script project; no data is shared across clients in Phase 1.
Security — Secrets	Twilio and Claude credentials shall be stored in Apps Script Script Properties, never in the Sheet or client-side/dashboard code.
Security — PII/PHI	No PII/PHI shall appear in URL query parameters; the inbound Twilio webhook uses POST only. Logged phone numbers are masked to the last 4 digits.
Security — Inbound auth	The public webhook shall require a shared-secret token, since Apps Script cannot read the standard Twilio signature header.
Auditability	Every inbound/outbound message and every escalation reason shall be appended to a per-patient log in the Sheet.

7. System Architecture Overview

The MVP is built on Google Apps Script bound to a Google Sheet, for the same reasons as any solo-practice tool needs to minimize new infrastructure: native Google Form→Sheet writes, cron-equivalent scheduling via time-driven triggers, a public webhook endpoint for inbound Twilio messages, and a private web-app endpoint for the staff-facing approval dashboard.

- Google Form (closed-category fields only) → Google Sheet (native write).
- Installable "on form submit" trigger → builds and sends first-contact SMS → logs status.
- Recurring time-driven trigger → scans Sheet for due reminders → sends and reschedules.
- Twilio inbound webhook (public deployment) → matches phone to patient row → local keyword filter → Claude API (logistics-only) drafts a reply, or the system escalates without calling the model.
- Staff dashboard (private deployment, Google-account gated) → approval queue → Approve & Send calls Twilio directly.

8. Data Requirements (Intake Sheet Schema)

Column	Field
A–E	Timestamp, Name, Phone, Reason for Visit, Preferred Appointment Window (from Form)
F	Status (pending / contacted / replied / reminder_sent / escalated / booked / closed)
G–H	LastContactAt, NextActionAt
I	ResponseTimeSeconds
J–K	AIDraftReply, ConversationLog
L–M	Approved (checkbox), ClientId

9. Assumptions & Constraints

- The clinic holds (or will create) their own Google Workspace, Twilio, and Anthropic accounts; credentials are not shared with or stored by the implementer.
- The local keyword filter is a best-effort MVP safeguard, not a compliance guarantee; it reduces but does not eliminate the chance of clinical text reaching the model.
- US SMS is subject to A2P 10DLC registration and STOP/HELP opt-out handling at the Twilio/carrier level.
- Apps Script has a per-project trigger quota (20), which is why reminders use one recurring sweep rather than one trigger per patient.

10. Risks & Mitigations

Risk	Mitigation
Clinical content reaches the AI system.	Two-layer defense: closed-category form fields (no free-text symptoms) plus a local keyword pre-filter that blocks the LLM call entirely on any match.
Real-patient PHI used before compliance controls are in place.	Explicit HIPAA warning at the top of this document and the setup guide; MVP is scoped as build/test only until BAAs are executed.
Public webhook abused to inject fake inbound messages.	Shared-secret token required on every inbound request; token is rotatable.

11. Success Metrics (MVP Acceptance Criteria)

- A test intake submitted via the Form receives a first-contact SMS in under 60 seconds.
- A logistics-only reply (e.g. "can I move it to the afternoon") produces a drafted AI response visible in the dashboard, unsent.
- A message containing clinical content (e.g. a symptom description) is escalated straight to staff, with the Apps Script execution log confirming the local keyword filter caught it before any Claude API call was made.
- Approving a logistics draft in the dashboard sends the SMS and updates the Sheet's status/approval fields.
- The dashboard accurately reflects status, last contact, next action, and average response time, with zero clinical fields present anywhere in the Sheet or dashboard.

12. Phase 2 (Out of Scope, Noted for Roadmap)

Execution of BAAs with Google Workspace, Twilio, and Anthropic to enable real-patient use; multi-tenant scaling by duplicating the Apps Script project + Sheet per client; optional migration to n8n/Make if trigger-quota ceilings are reached.